



Factorisation

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Division of Algebraic Expressions

We have learnt how to add and subtract algebraic expressions. We also know how to multiply two expressions. We have not however, looked at division of one algebraic expression by another. This is what we wish to do in this section.

We recall that division is the inverse operation of multiplication. Thus, $7 \times 8 = 56$ gives $56 \div 8 = 7$ or $56 \div 7 = 8$.

We may similarly follow the division of algebraic expressions. For example,

(i) $2x \times 3x^2 = 6x^3$

Therefore, $6x^3 \div 2x = 3x^2$

and also, $6x^3 \div 3x^2 = 2x$.

(ii) $5x(x + 4) = 5x^2 + 20x$

Therefore, $(5x^2 + 20x) \div 5x = x + 4$

and also $(5x^2 + 20x) \div (x + 4) = 5x$.

We shall now look closely at how the division of one expression by another can be carried out. To begin with we shall consider the division of a monomial by another monomial.

Division of a monomial by another monomial

Consider $6x^3 \div 2x$

We may write $2x$ and $6x^3$ in irreducible factor forms,

$$2x = 2 \times x$$

$$6x^3 = 2 \times 3 \times x \times x \times x$$

Now we group factors of $6x^3$ to separate $2x$,

$$6x^3 = 2 \times x \times (3 \times x \times x) = (2x) \times (3x^2)$$

Therefore,

$$6x^3 \div 2x = 3x^2.$$

A shorter way to depict cancellation of common factors is as we do in division of numbers:

$$77 \div 7 = \frac{77}{7} = \frac{7 \times 11}{7} = 11$$

Similarly,

$$\begin{aligned} 6x^3 \div 2x &= \frac{6x^3}{2x} \\ &= \frac{2 \times 3 \times x \times x \times x}{2 \times x} = 3 \times x \times x = 3x^2 \end{aligned}$$

Division of a polynomial by a monomial

Let us consider the division of the trinomial $4y^3 + 5y^2 + 6y$ by the monomial $2y$.

$$4y^3 + 5y^2 + 6y = (2 \times 2 \times y \times y \times y) + (5 \times y \times y) + (2 \times 3 \times y)$$

(Here, we expressed each term of the polynomial in factor form) we find that $2 \times y$ is common in each term. Therefore, separating $2 \times y$ from each term. We get

$$\begin{aligned} 4y^3 + 5y^2 + 6y &= 2 \times y \times (2 \times y \times y) + 2 \times y \times \left(\frac{5}{2} \times y\right) + 2 \times y \times 3 \\ &= 2y (2y^2) + 2y \left(\frac{5}{2} y\right) + 2y (3) \\ &= 2y \left(2y^2 + \frac{5}{2} y + 3\right) \text{ (The common factor } 2y \text{ is shown separately.)} \end{aligned}$$

Therefore, $(4y^3 + 5y^2 + 6y) \div 2y$

$$= \frac{4y^3 + 5y^2 + 6y}{2y} = \frac{2y(2y^2 + \frac{5}{2}y + 3)}{2y} = 2y^2 + \frac{5}{2}y + 3$$

Alternatively, we could divide each term of the trinomial by the monomial using the cancellation method.

$$\begin{aligned} (4y^3 + 5y^2 + 6y) \div 2y &= \frac{4y^3 + 5y^2 + 6y}{2y} \\ &= \frac{4y^3}{2y} + \frac{5y^2}{2y} + \frac{6y}{2y} = 2y^2 + \frac{5}{2}y + 3 \end{aligned}$$

Here, we divide each term of the polynomial in the numerator by the monomial in the denominator.

Division of Algebraic Expressions Continued (Polynomial $\div$ Polynomial)

- Consider $(7x^2 + 14x) \div (x + 2)$

We shall factorise $(7x^2 + 14x)$ first to check and match factors with the denominator:

$$\begin{aligned} 7x^2 + 14x &= (7 \times x \times x) + (2 \times 7 \times x) \\ &= 7 \times x \times (x + 2) = 7x(x + 2) \end{aligned}$$

Will it help here to divide each term of the numerator by the binomial in the denominator?

Now $(7x^2 + 14x) \div (x + 2) = \frac{7x^2 + 14x}{x + 2}$

$$= \frac{7x(x + 2)}{x + 2} = 7x \quad (\text{Cancelling the factor } (x + 2))$$